Laboratory-Scale Distributed Pressure Measurements of Blade Interaction with JSC-1A Lunar Simulant

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Abstract: Because of the requirement for high reliability and very low weight, extraterrestrial excavator blade design for in situ resource utilization (ISRU) must be more precise than for current terrestrial practice. Three-dimensional finite-element and discrete-element models should be employed to enhance the precision of designs. These models require that the pressure distribution across the blade be known; however, this information is not available in the literature. To provide the information, this project measured the pressure distribution on a suite of laboratory-scale blades. The project constructed a suite of laboratory-scale model blades, designed and implemented distributed pressure and total load measurements, developed soil preparation techniques, measured pressure distributions and errors, and created empirical pressure distribution models. The models include: an array of point pressures, a quadratic function, and an exponential function. The work concluded that a V-shaped blade with a tapered curve is the best configuration, and multiple passes of small-depth cuts is the best excavation operating procedure. **DOI: 10.1061/(ASCE)AS.1943-5525.0000224.** © *2013 American Society of Civil Engineers*.

CE Database subject headings: Lunar materials; Excavation; Space construction; Load factors; Measurement; Simulation.

Author keywords: Lunar materials; Excavation; Space construction; Soil-blade interaction; Blade-distributed pressure measurement; Extraterrestrial excavation.

Introduction

If outposts are developed on extraterrestrial bodies, such as the moon, several soil handling tasks are required, such as creating roads and landing sites (Mueller and King 2007). Outpost development requires a robotic machine, like the National Aeronautics and Space Administration prototype shown in Fig. 1, to build berms, grade and compact roads, excavate foundations, smooth landing pads, and excavate trenches for utilities (Mueller et al. 2009). Robotic excavators must be designed to withstand the stresses of excavation without failing. Relatively large design factors are applied to overdesign terrestrial excavators for operator abuse, long life, and varied conditions, resulting in equipment power and sizes that range from 13 to 382 kW and 1,558 to 84,980 kg. Launch costs of such large equipment are prohibitive, and therefore precise design must take the place of design factors. Precise design begins with analysis of the pressures on the blade. If excavation pressures are known, finite-element and discrete-element models can be used to calculate stresses and displacement in the frame and joints, determine tractive requirements, size motors, and provide data for trade studies.

Overdesign would consider a worst case scenario where the corner of the blade impacts a large rock, which might be buried and unseen (Fig. 1). Finite-element analysis (FEA) for this scenario where a 22,241-N point load was applied to simulate the corner of the blade hitting a large rock buried in soil at full speed resulted in a factor of safety of 0.013 for the pathfinder design shown in Fig. 1. The FEA was verified when the corner of the blade hit a wet pile of sand when the vehicle was traveling at full speed and the blade support buckled.

A sophisticated robotic excavator would sense the increased stress from a worst case scenario and react by pulling back quickly. Consequently, a design should consider the normal operating load as well as the worst case scenario. The average operating load can be simulated with traditional models (King et al. 2011). However, these models are two-dimensional, or if they are three-dimensional, they provide only average uniform pressure distributions over the blade surface, which doesn't accurately simulate practice or take full advantage of the capability of the FEA software (King et al. 2010). Nonuniform load distribution is typical of excavation practice, because soil doesn't press against the upper corners, the surcharge shape is curved, the middle of the blade pushes the most soil, and the surcharge extends beyond the edge of the blade (Fig. 2). This project measured nonuniform pressures on laboratory-scale blades and developed load models that will improve the accuracy of extraterrestrial excavator blade design.

Measurements

Pressure was measured on a suite of laboratory-scale straight and V-shaped blades (Fig. 3) (Brewer 2011). The straight blade simulates the LANCE-type blade. The V-shaped blade design may be better for extraterrestrial application, because it minimizes the load during excavation by diverting the soil to either side of the blade. Two sets of straight blades compare blade widths (20.32, 25.40, and 30.48 cm) and face curvature, while the V-blades compare the blade shapes as well as angles (15, 30, and 45°). The following were varied by evaluating change in one variable at a time while keeping all others at a base configuration: the material being excavated, the cutting depth, the cutting angle, the rake angle, and the blade width (Table 1). Additional tests were performed to compare straight and curved blades (Table 2). Some measurements were made with

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Note. This manuscript was submitted on September 2, 2011; approved on March 6, 2012; published online on March 8, 2012. Discussion period open until June 1, 2013; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Aerospace Engineering***, Vol. 26, No. 1, January 1, 2013. ©ASCE, ISSN 0893-1321/2013/1-105−116/$25.00.

Fig. 1. National Aeronautics and Space Administration's chariot rover with attached pathfinder blade

Fig. 2. Nonuniform loading on the pathfinder blade

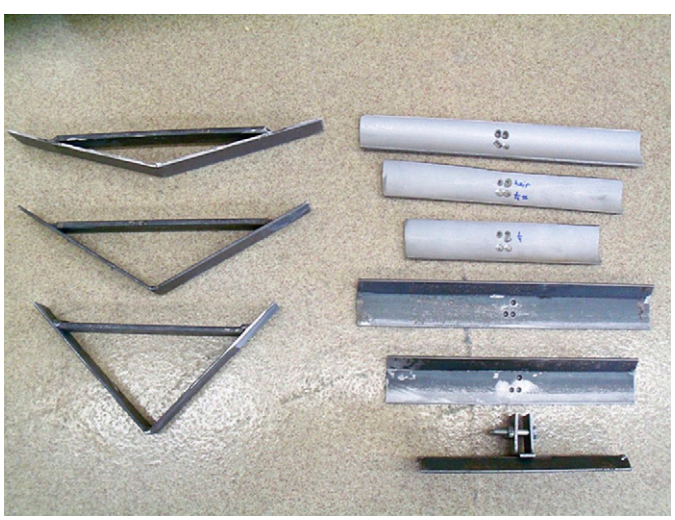

Fig. 3. Model blade suite

a square rod instead of a blade in order to compare with previous measurements by Gefreh (2008) to evaluate repeatability and error.

The blades were mounted in the Colorado School of Mines (CSM) soil excavation measurement apparatus (Fig. 4), where a soil tray moves under an excavation tool (Johnson and King 2010). The National Instruments PXI data acquisition (DAQ) and control system measures the velocity and horizontal and vertical forces automatically, maintains constant velocity, and stops the test automatically at the end of travel.

The traditional test bed DAQ measures total load on the cutting tool, and therefore a Tekscan *I-Scan* universal serial bus (USB) evolution single-input pressure mapping system (Tekscan 2009) was added to measure distributed pressure over the surface of the tool. The system consists of a Model 6300 thin-film resistive sensor and a USB computer interface with data acquisition and analysis software. The sensor was attached to the face of the blade being tested, and the USB interface was mounted to the outer carriage (Fig. 5). This system is typically used for analyzing the contacts between two surfaces that form a seal and have never been applied to small-scale excavation pressure measurements prior to these tests. The thin-film resistive sensor has an active measurement area of

Table 1. Base Configuration and Range for Test Sets by Variable

Variable	Base	Range
Material	JSC-1A	JSC-1A, Ottawa Sand
Cutting depth (cm)	0.635	0.127–0.635 (0.127 increments)
Rake angle (degrees)	0	0, 8, 12.5
Blade angle (degrees)	0	0, 15, 30, 45
Blade width (cm)	30.48	20.32, 25.40, 30.48
Blade curvature	Flat	Flat, curved

Table 2. Base Configuration and Range for Additional Curved Blade Test Sets

Variable	Base	Range
Material	JSC-1A	—
Cutting depth (cm)	0.635 cm	0.127–0.635 (0.127 increments)
Rake angle	—	—
Blade angle (degrees)	0	—
Blade width (cm)	30.48	20.32, 25.40, 30.48
Blade curvature	Curved	—

Fig. 4. Soil excavation measurement apparatus

26.42×3.35 cm, a spatial resolution of 25.8 sensing cells (sensels) (per cm^2), a maximum sampling rate of 100 Hz, and a pressure range of 0.00−68.95 kPa that can be adjusted for increased resolution. The sensor is made of two, thin, flexible polyester sheets with conductive electrodes deposited on the inner surface of each sheet. The inner surface of one sheet forms a row pattern, while the inner surface of the other forms a column pattern. The intersection of each row and column creates a sensel. The vertical spacing is 13.0 sensels/cm, while the horizontal spacing is 2.0 sensels/cm. The higher vertical resolution was desired for precise vertical load profile measurements. Electronics scan the sensels and acquire a resistance value at each sensel. The device converts the measurements from analog to digital with 8-bit resolution and communicates them to the computer via USB. The response time for the system to acquire the sensel data and communicate it to the computer was 15.2 ms. The *I-scan* (Tekscan 2009) data acquisition software running on the computer provides low-level sampling control, sensitivity adjustment, and signal processing. The sampling rate can be set between 0.0153 and 100 Hz with 100 Hz used for all data in this work. The pressure range can be set from 69 to 68,950 kPa. Ideally, the low-pressure range would be used for higher-resolution measurements, but it was set higher than desired because of the calibration method used.

Soil Properties

Two soils were tested: Ottawa Sand and JSC-1A Lunar Soil Simulant. Ottawa Sand was chosen as a base for comparison with tests performed by Gefreh (2008). Ottawa Sand is a pure silicate with a very uniform grain size and subangular grains. The grain distribution and subangular particles make it resistant to settling and compaction, which enhances test repeatability. Ottawa Sand has an internal angle of friction of 33° and a cohesion value of 3.520 kPa (King and van Susante 2007).

JSC-1A mimics the material properties of lunar regolith. It is made from crushed basalt, and therefore the particles are very angular, and the particle size distribution is much more dispersed than Ottawa Sand. JSC-1A is denser than Ottawa Sand. The internal angle of friction, cohesion, external angle of friction, and adhesion are 37°, 1.793 kPa, 23°, and 39 Pa, respectively (King and van Susante 2007). These characteristics make the simulant highly prone to settling and compaction, similar to lunar soil (Carrier et al. 1991). Therefore, soil preparation prior to each test is extremely important.

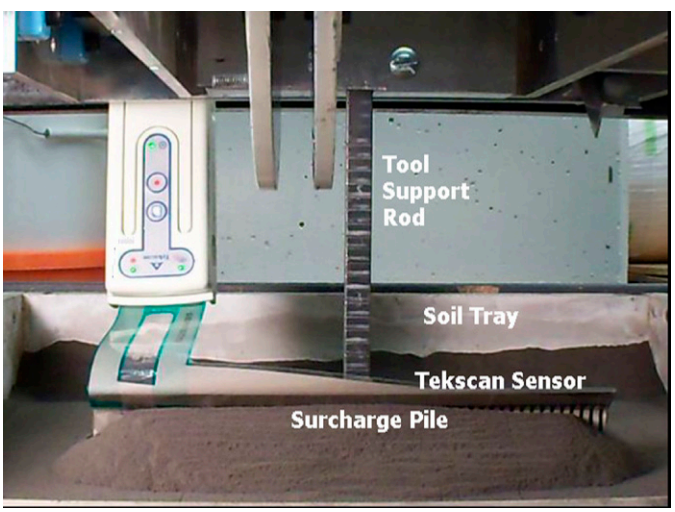

Fig. 5. Tekscan sensor and blade mounted on the CSM test bed

Calibration

Both the vertical and horizontal load cells were calibrated with hanging precision weights at three different times during the project: at the outset of the project, after a failure of the horizontal load cell when the 222-N vertical load cell was used, and after the horizontal load cell and amplifier were replaced. The calibration relationships are summarized in Table 3.

Tekscan provides an equilibration/calibration device, but it was not included in the Tekscan rental system, and therefore the sensor was calibrated with precision weights. Load was distributed with a $2.54 \times 15.24 \times 0.635$ cm steel plate placed on the sensor. Two loads were applied, the mass of the steel plate (217.9 g) and 500 g of weight on the plate. Table 4 presents the calibration results.

Measurement Procedure

The testing procedure was designed to reduce error while performing the 250 tests as quickly and efficiently as possible. The soil was prepared in the same fashion for both Ottawa Sand and JSC-1A Simulant in order to reduce compaction that would increase after each test and greatly affect the measured forces by raking the soil with a hand-held garden claw backward and forward. The soil was leveled by mounting a plexi-glass panel and moving the tray under the panel. The panel was mounted at the same height for every test, ensuring the same average density and average compaction for all tests.

The Tekscan sensor was attached to the blade face with removable spray adhesive to facilitate mounting on a series of blades. The bottom right corner of the sensing area was aligned with the bottom right corner of the blade. Excess sensor material was cut from the right side to eliminate error from pushing soil with the sensor and to avoid boundary-effect errors with the sides and bottom of the box and to place the sensels as close to the edge of the blade as possible. The sensor does not cover the entire blade face, and therefore the analysis only considers data for the right half of the blade.

The cutting depth was set by raising or lowering the tool support rod in 0.127-cm increments over the range of 0.127−0.625 cm.

Because the Tekscan sensor needed to be cut to fit on the smaller width blades, the wider 30.48-cm flat blade was tested first in the Ottawa Sand. Then it was tested in the JSC-1A over the ranges of

Table 3. Summary of Calibration Relationships

Relationship	First calibration	Second calibration	Third calibration
Horizontal calibration (N)	$F = 36.532V - 27.224$	$F = 79.676V - 270.61$	$F = 38.433V - 37.429$
Horizontal coupling (V)	$V = \begin{cases} 0.0012F + 5.8603 & \text{if} \quad F < 0 \\ -0.0016F + 5.8565 & \text{if} \quad F > 0 \end{cases}$	Not applicable	$V = \begin{cases} -0.0002F + 5.6329 & \text{if} \quad F < 0 \\ -0.0007F + 5.6263 & \text{if} \quad F > 0 \end{cases}$
Vertical calibration (N)	$F = -66.767V + 394.18$	Not applicable	$F = -76.737V + 424.63$
Vertical coupling (V)	Not measured	Not applicable	$V = -0.00002F + 0.9973 \quad F < 0$

cutting depths and rake angles. Then, the test suite was repeated with the 30.48-cm curved blade. Next, the V-blade tests were performed. After that, the sensor was cut to fit on the 25.40-cm blades, and the tests were repeated. Finally, the sensor was cut to fit the 20.32-cm blades, and the tests were repeated. The soil tray accelerated to 200 mm/min and maintained that rate before decelerating at the end of travel for all of the 250 tests performed.

Most of the Tekscan data were processed in *Mathematica* (*Mathematica 7*), but some color contour plots were produced using the Tekscan software. Distribution plots were imported into *Mathematica* for use in developing the distributed pressure models. Data for the V-blades was imported into *Microsoft Excel* (*Microsoft Excel*) and converted to horizontal data. The load cell data were reduced in *Microsoft Excel* by averaging over five data points. *Microsoft Excel* and *Mathematica* were used to generate plots and analyze errors.

General comparisons between blades are based on data at 33 s, the time when the surcharge reaches the top of the sensing area for the base tests.

Results

Fig. 6 presents a typical graph that compares the total load measured by the Tekscan sensor and the load cell. Results from all of the scenarios are presented in Brewer (2011). The results from the Tekscan sensor and the load cells are within the maximum expected error.

Table 4. Tekscan Sensor Calibration Constants

Sensor	Scale Factor A	Exponent B
1	0.0467	0.692
2	0.0347	0.703

Fig. 7 shows the measurement data distribution of the base tests in a color contour plot. The total load for this modified distribution was calculated as 2.26 N for half of the blade, which compares favorably with the average load cell measured value of 2.21 ± 0.47 N.

Fig. 8 presents the pressure distribution results, averaged for all 25 base tests. Fig. 9 shows the results of smoothing with a 5×5 sensel mask.

The average load versus cut depth results for flat and curved blades can be approximated with a second-order polynomial (Fig. 10). Because high loads mean a stronger and heavier machine with larger motors requiring higher power, lightweight machines for extraterrestrial application should excavate thin cuts and take multiple passes if necessary.

The data suggest that wider blades are preferred, because the increase in force is the smallest going from the 25.40-cm blade to the 30.48-cm blade (Fig. 11). However, wider blades increase the weight of extraterrestrial machines with the weight of the blade; more importantly, the moment with uneven loading increases the stress in the blade supports and machine frame, increasing the overall weight of the machine. Consequently, the width should be limited to that necessary to cut clearance for the machine to travel behind the blade.

Higher rake angles reduce the excavation forces (Fig. 12). This can be achieved practically with a curved blade. The excavation width decreases as the angle increases; therefore, longer blades are needed at higher angles to cut clearance for the machine. The transverse horizontal pressure distribution cancels out for V-blades

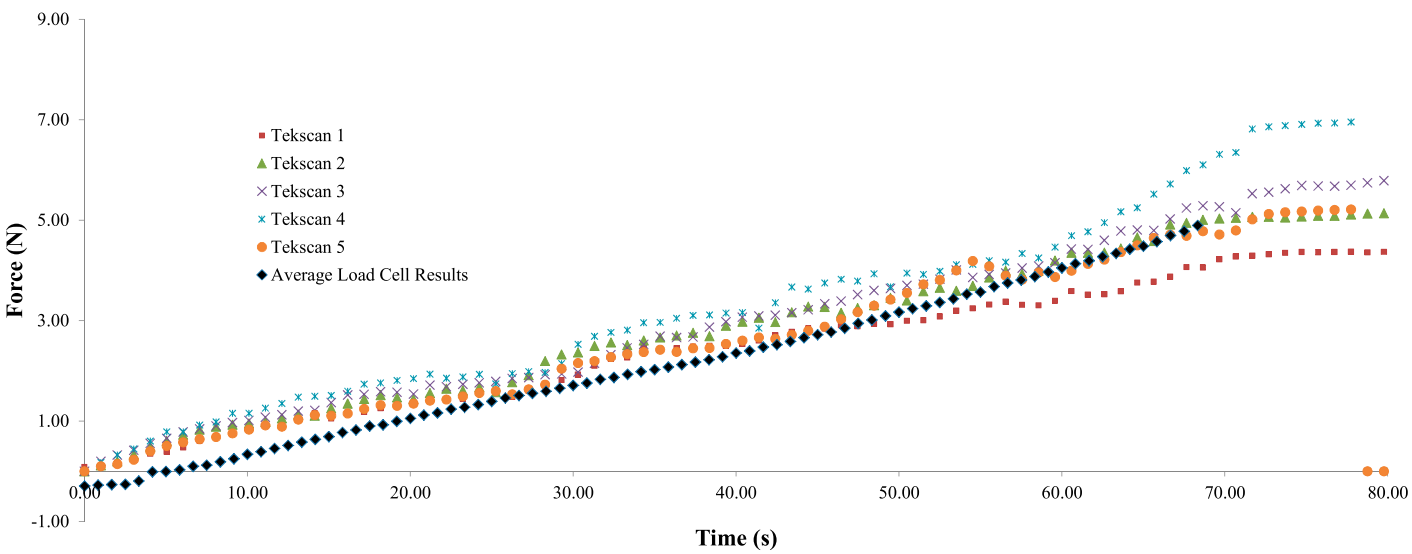

Fig. 6. Tekscan average load measurements compared with average load cell results with a 25.4-cm flat blade at 0.635-cm cut depth

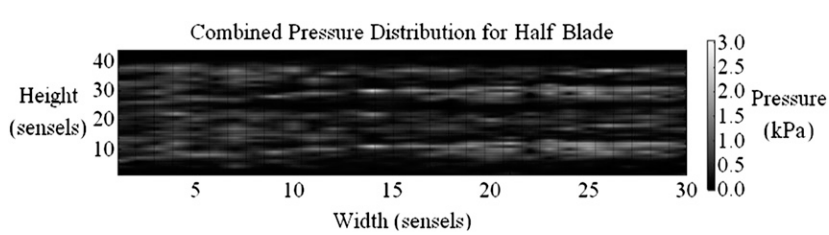

Fig. 7. Color contour plot of the half-blade combined base test distribution; blade center at width = 0

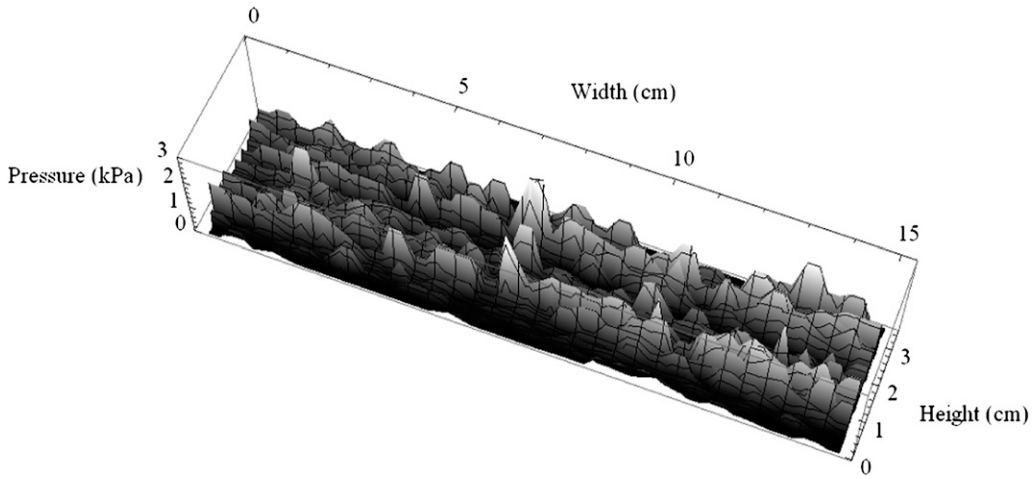

Fig. 8. Average combined pressure distribution for base tests modified by superposition; blade center at width = 0

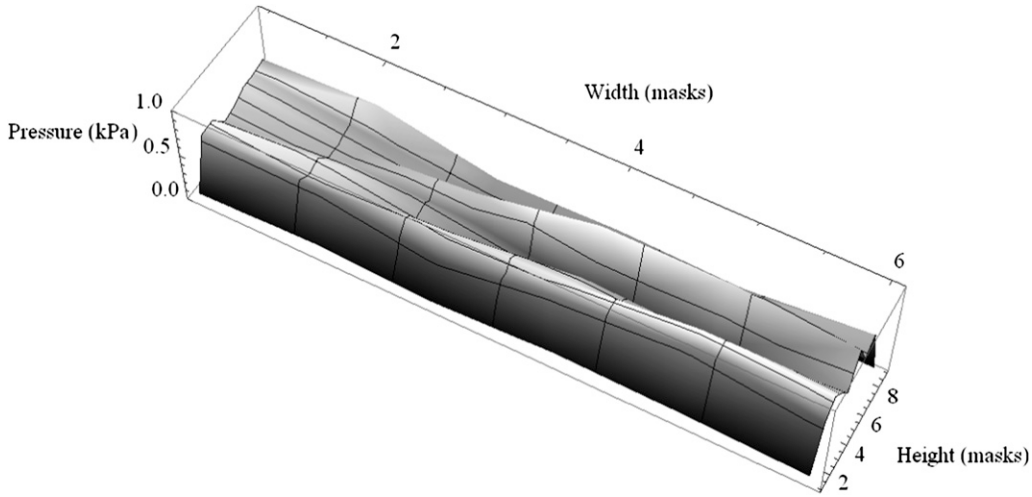

Fig. 9. Pressure distribution for base tests at 33 s smoothed with a 5 × 5 sensel mask; blade center at width = 0

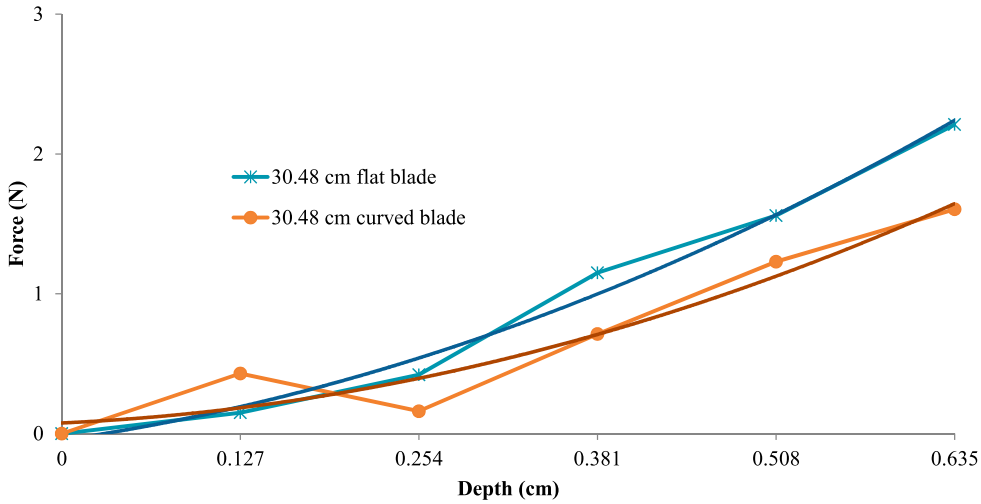

Fig. 10. Load versus cutting depth

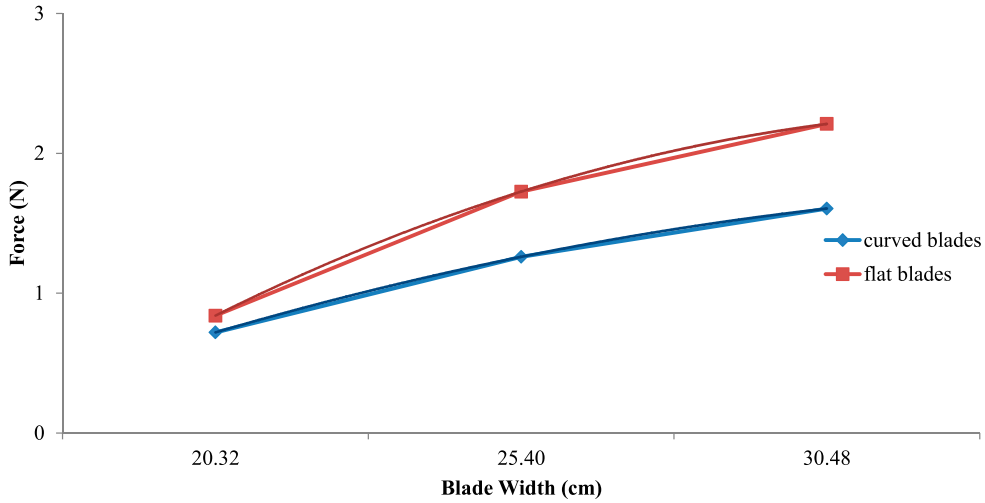

Fig. 11. Load versus blade width

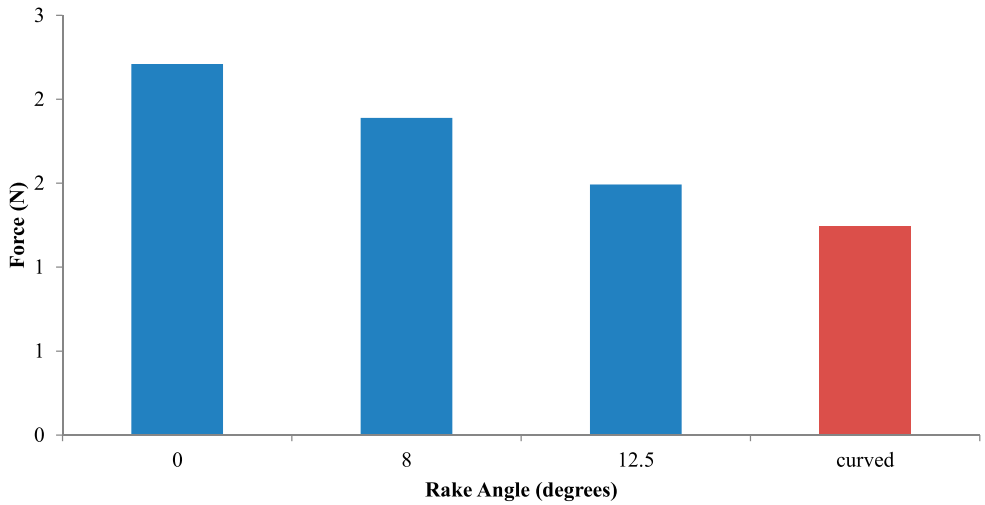

Fig. 12. Load versus rake angle

when cutting uniformly across the blade width, but not for angled blades. However, the V-blades require more material and would increase the overall mass. Assuming uniform loading across the blade width, a V-blade or angled blade is better than a straight blade when considering only the resulting pressures in the direction of travel (Fig. 13). When considering the transverse horizontal pressure distribution, the V-blade becomes the most favorable blade shape, because the pressure distributions for the two halves are equal and opposite. The greatest change in force occurs between the 30 and 45° blade angles, meaning the optimal angle should be between 30 and 45°. The 30° angle provides the widest blade. However, a detailed evaluation of the mass and reliability of the V-blade is recommended. With precise design and advanced controls, the decrease in reliability would be minimal. The V-blade was also favored, because the surcharge that accumulates doesn't build up in the center, similar to flat blades, which facilitates a lighter machine.

Discussion

The cutting depth precision was limited to about a one-quarter turn of the height adjustment rod. This error is equivalent to ±0.03 cm. The bottom corner of the Tekscan sensor could only be lined up with the bottom corner of the blade face with a precision of 0.07 ± 0.07 cm from either edge.

An air bubble that measured 1.35 cm from top to bottom at 33 s was caused by bending the sensor to fit the model blade faces (Fig. 14). It occurred on all blade geometries. As the surcharge climbed on the blade, the air bubble was pushed toward the top of the sensing area causing pressures to accumulate along the top edge and to lessen under the bubble. To remove this effect from the results, the distribution data from the top portion of the base tests was replaced with data acquired at 20 s, before the surcharge pressure measurement was affected by the air bubble (Figs. 15 and 16).

A step decrease was applied to the sensor, and the response was measured over time to study the dynamic response of the Tekscan system. The output exhibited a first-order response from which the time constant was found to be 15.2 ms. The manufacturer confirmed that the sensor response could be modeled with roughly first-order behavior, but the manufacturer provided a significantly smaller time constant of 20 μs.

To evaluate the dynamic response, a soil test was completed while recording values from both the Tekscan sensor and the horizontal load cell. With the time constant known, the expected output

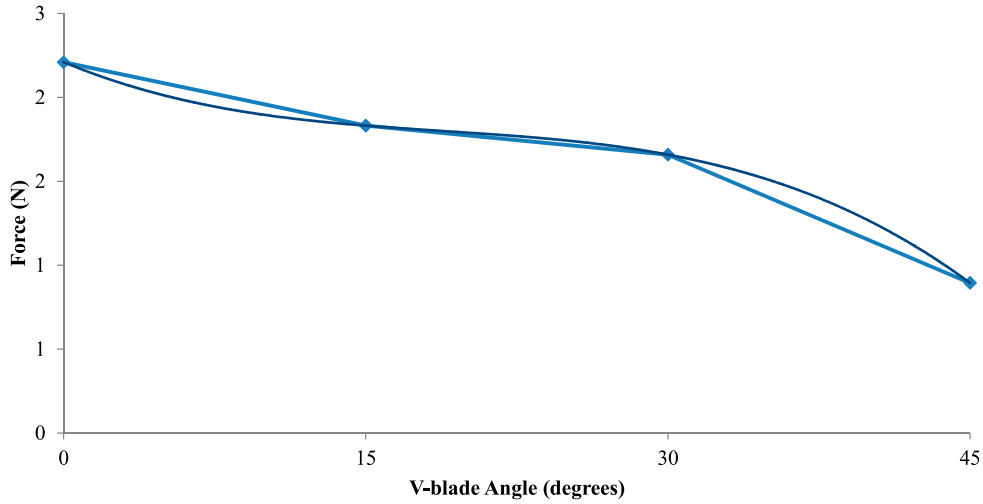

Fig. 13. Average half-blade load results versus blade angle

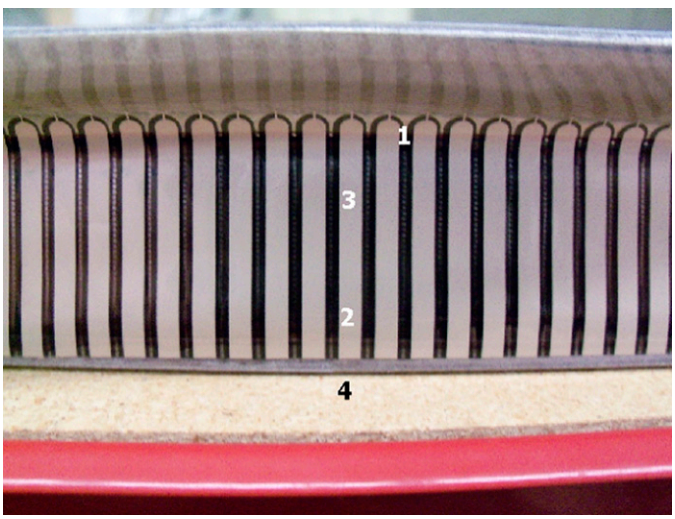

Fig. 14. Tekscan sensor mounted on a flat blade showing: (1) bending; (2) air bubble initial; (3) final positions; (4) alignment error

load values were calculated from the known input values using the convolution, $y(t) = h(t)^{*}x(t)$, where y is the output signal, $h(t) = (1/\tau)e^{-(t/\tau)}$, x is the input signal, t is time, and τ is the time constant. The calculated output signal is equivalent to the input signal with less noise, where the load cell time step was increased to 0.017 s, and the Tekscan time step was set to 0.010 s. The convoluted output signal had the same shape as the Tekscan half-blade load data, indicating that the 100-Hz sampling rate was adequate.

Five tests were conducted to characterize noise introduced by the physical test assembly (e.g., vibration) and electrical components. The test group varied cut depth from 0 to 0.635 cm, advance rate from 100 to 500 mm/min, clean and dirty tools, and square rods and flat blades. Noise was measured about a baseline voltage, and there was no determinable change in power-spectral density across the frequency range of 20–150 Hz; the Nyquist frequency for measurements was taken at 300 Hz. A Gaussian noise distribution was assumed for further analysis. The minimum signal-to-noise ratio, the output voltage divided by the standard deviation of the signal, was approximately 126 for a fully loaded blade (load cell voltage = 773 mV) running at a high rate of advancement (500 mm/min). The arrangement of the sliding bearings in both the vertical and horizontal directions on the test assembly minimizes the effects of vibration, and therefore the nonuniform motion of the test bed was the primary cause of mechanical noise. Sand accumulating on the worm gear might cause uneven horizontal motion. Consequently, the gear and associated components were cleaned and greased thoroughly before the tests, reducing the standard deviation by more than 1.3 mV. The remaining noise in the signal was assumed to be electrical, mainly as a result of magnetic induction from concurrent wiring and amplifier noise. The noise analysis results for both load cells and amplifiers varied from 0.005 to 0.016 V.

The load was not evenly distributed during calibration, resulting in pressure concentrations causing the measurement range to be set too high, reducing resolution. The maximum pressure on one sensel never exceeded 3.8 kPa, and the average pressures were about 10% of the pressure range. The pressures measured from all of the sensels were averaged over the duration of the test, producing about 0.45 kPa. The sensor resolution was about 6% of the average pressure. The resolution was 0.059 kPa, and the error was ± 0.027 kPa.

Random variations were investigated in two ways: variations in the pressures measured by each Tekscan sensor sensel and variations in the total load data from the Tekscan sensor and the load cells. The random variation between tests was characterized by calculating the standard deviation of the pressure values at each sensel and plotted. The maximum standard deviation for one sensel between tests was found to be 2.28 kPa, and the average standard deviation was found to be 0.52 kPa. These values are high for an average measured pressure of 0.45 kPa, and are probably caused by natural variation in soil density. Random variation between sensels was characterized by evaluating the average pressure distribution through 5×5 sensel resolution masks. For each 5×5 resolution mask, the pressures from the 25 sensels under the mask were averaged together and the standard deviation calculated. The error between sensels was less than the sensel error between tests. From these distributions, the maximum standard deviation between sensels was 0.74 kPa, and the average standard deviation was 0.37 kPa. The highest variations were in the high-load areas, as expected.

A measurement error could be introduced in the load cell data from coupling between the horizontal and vertical loads. There are two parts to this error: a constant offset and the sensitivity of vertical load to horizontal load variation. Even though the test bed frame was

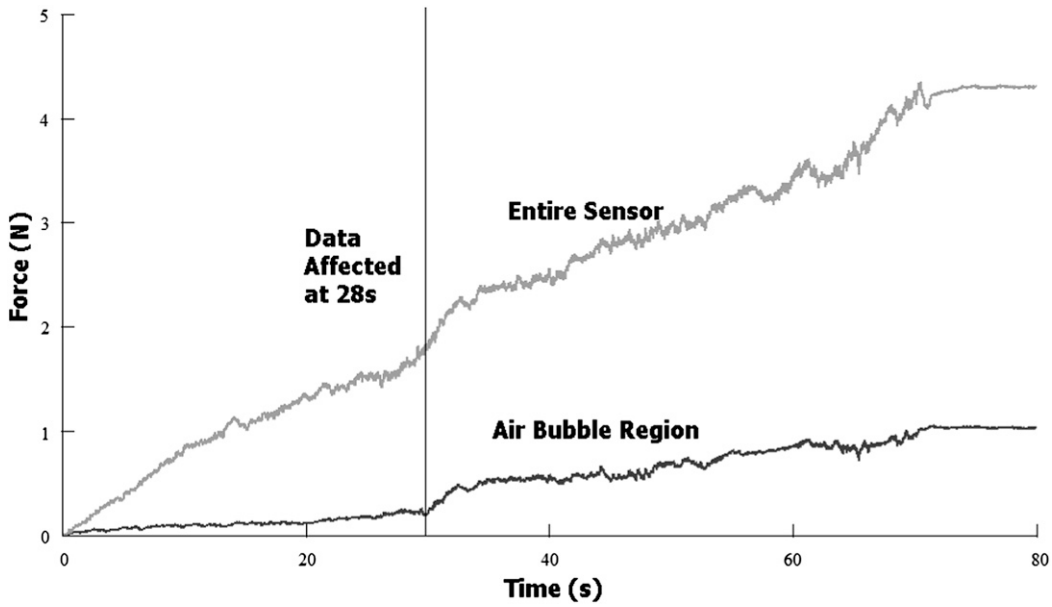

Fig. 15. Average force vs. time graph for air bubble region and entire sensor area; Tekscan measured pressure distribution at 28 seconds; blade center at $x = 0$

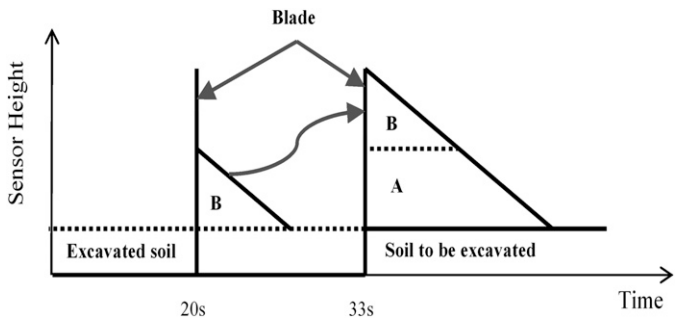

Fig. 16. Cross section sketch of surcharge superposition regions (region A at 33s combined with region B at 20s)

very stiff to minimize this problem, the linear bearings allow small angular movement to achieve smooth motion, which allows the carriage to tilt slightly under load that creates a moment and residual force. The coupling because of a vertical load was measured with calibrated weights to be nearly zero, and therefore the apparent coupling was most likely because of the calibration setup. If the force was not purely vertical or horizontal, then some force would register on the other. This error was accounted for in random variation.

The random variation includes the coupling, soil compaction, and sensor mounting. It was determined by calculating the standard deviation between tests of the data from tests with the largest blade with the largest expected load. The maximum random variation in the load cell data was ±0.23 N, and in the Tekscan data were ±0.57 N. The combined data has a standard deviation of ±0.47 N.

Square rod tests were conducted to replicate the measurements by Gefreh (2008). Tables 5 and 6 compare the square rod results. The standard deviation accounts for the differences with one exception. The cause for the difference could be differences in soil preparation or soil box width. If the width affects the results, then the JSC-1A tests would have lower forces in a wider sand box. Because of low adhesion, Ottawa Sand forces should not change much with a wider box. Because the JSC-1A Simulant was exposed to the atmosphere for several months, the soil properties (particularly moisture) could

Table 5. Square Rod Force Result Comparison in Ottawa Sand

Depth (cm)	Test	Average force (N)	Difference in force (N)	Standard deviation of force (N)
0.6350	Brewer	0.1266	0.1573	0.2223
	Gefreh	0.2839		0.0237
2.5400	Brewer	0.6519	0.9663	0.2561
	Gefreh	1.6182		0.0867
3.8100	Brewer	3.5295	0.1846	0.3432
	Gefreh	3.3449		0.1331

Table 6. Square Rod Force Result Comparison in JSC-1A Simulant

Depth (cm)	Test	Average force (N)	Difference in force (N)	Standard deviation of force (N)
0.6350	Brewer	0.5487	0.1273	0.1354
	Gefreh	0.4214		0.0474
1.2700	Brewer	0.7513	0.0853	0.1300
	Gefreh	0.8366		0.0589
3.8100	Brewer	2.7030	3.0939	0.3425
	Gefreh	5.7969		0.3352

have changed, resulting in different measured forces. The change in force between soils was within the standard deviation of Gefreh's (2008) JSC-1A tests, and therefore change in soil properties was not a major influence.

The average Tekscan results agree with the load cell results in all but two test sets. In the Ottawa Sand test set, the Tekscan results had unusually low force values. The low forces were most likely from using a sensor that had not been adequately conditioned, because these were the first tests performed. For this reason, only the load cell data were considered for the Ottawa Sand tests. The tests performed using V-shaped blades, which have excessively high-load cell results that are the result of added moment in the measurement because of the shape of the V-blades and the position of the mount. The error between measurement systems increases as the V-blade angle increases. An increased V-blade angle

corresponds to a longer distance between the tip of the V-blade and the blade mount.

The surcharge reaches the top guard of the flat blade after running for 33 s, or after traveling 11 cm. Steady state cannot be reached within 50 cm of travel, the upper limit of the soil tray's testable range of motion. To reach steady state, the test bed would have to be modified to increase the soil tray's range of motion.

Distributed Pressure Models

The distributed pressure models were developed in *Mathematica* using the least-squares approach for an average of 25 base tests at 33 s, when the surcharge reaches the top of the sensor. It is expected that extraterrestrial robotic equipment control would limit operations to that scenario. Three types of distributed pressure models were developed: direct input to an FEA model of an array of point pressures, a quadratic model specifically for input into the SolidWorks Simulation FEA distributed pressure option, and an exponential model for comparison with the quadratic model, as well as for input into other FEA programs. The exponential and quadratic models use an evenly distributed 0.23-kPa pressure in the cutting region, which extends the width of the blade and from the bottom of the cut to the cut depth d that is limited to 0.635 cm.

The point-pressure model distribution is an array of pressures for each sensel of an 0.08×0.51 cm area on the blade face (Fig. 17). The data for this model was smoothed using a 2-point moving average function in both the horizontal and vertical directions. The uncertainty values, shown by dots, are the average sensel error between tests (0.52 kPa).

Fig. 18 shows the quadratic pressure model produced for the 4.74-N worst case target load, as well as the model overlaid with the point-pressure model data. The model was developed by scaling the least-squares fit of the surcharge data found in *Mathematica* by 0.502, and using the average pressure (0.23 kPa) in the cutting region. To be very conservative, the quadratic model of cutting pressure uses the upper limit of error (0.46 kPa). The surcharge distribution was found by scaling the least-squares fit by 1.06 to produce a total load of 4.74 N. This scaling factor shows that the upper-limit fit is close to the original least-squares fit. The quadratic pressure model with a target load of 4.74 N is of the form

$$P_{\text{surcharge}} = 0.42796 - 0.06647x^2 + 0.25170x - 0.00854xy - 0.00194y + 0.00187y^2$$

$$P_{\text{cut}} = 0.46$$

where $P_{\text{surcharge}}$ and P_{cut} = pressures in kilopascals, x = surcharge height in centimeters, starting at the bottom of the surcharge region, and y = distance along the blade width in centimeters, starting at the center of the blade. The P_{cut} is constant over the cutting region. As with the exponential model, this distributed pressure model can be

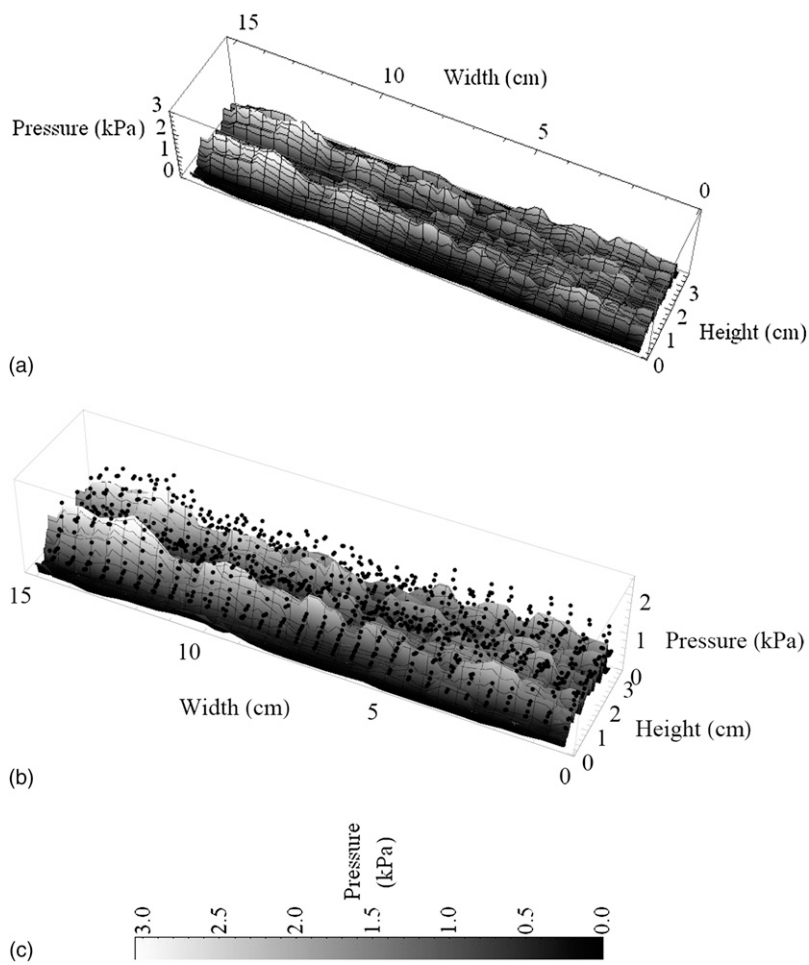

Fig. 17. (a) Average point-pressure distribution model with blade center at width = 0, cut height at 0.635 cm; (b) overlay of uncertainty values represented by dots; (c) legend for pressure values

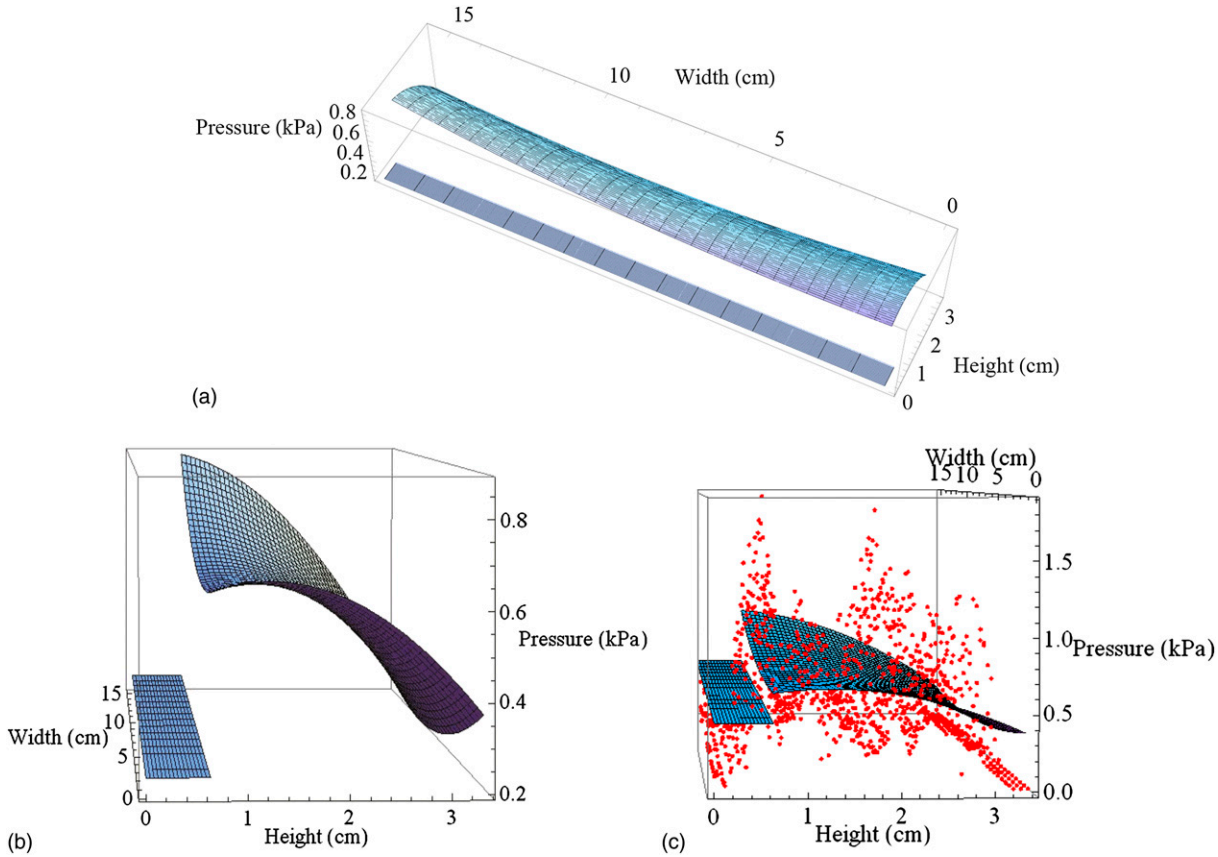

Fig. 18. (a) Quadratic pressure model for base test scenario for 4.74-N target load with blade center at width = 0, cut height at 0.635 cm; (b) side profile; (c) overlay of point pressure model data represented by dots

applied in the opposite y-direction to form a complete blade distribution. For the surcharge region in both the quadratic and exponential models, the allowable ranges for x and y are 0.00 − 2.72 cm and 0.00 − 15.24 cm, respectively. For the cutting region, the height is 0.635 cm in both the quadratic and exponential models.

The error in the pressure models caused by the 0.07-cm uncertainty in the sensor position was found using the relationship $\delta P = (\partial P/\partial x)\delta x + (\partial P/\partial y)\delta y$, where δP is the resulting error, and both δx and δy are equal to the uncertainty in the sensor position. For the quadratic pressure model with a target load of 2.25 N, the error was found to be $\delta P = 0.00820 - 0.00465x - 0.00040y$.

The error for the quadratic pressure model with a target load of 4.74 N was found to be

$$\delta P = 0.017605 - 0.00990x - 0.00086y$$

The exponential pressure models were generated in the same way as the quadratic models using the least-squares fit method in *Mathematica*, then scaling the model by 1.08 to produce the 4.74-N worst-case load. The exponential model curves less than the quadratic models and in the opposite direction (Fig. 19).

The exponential pressure model with a target load of 4.74 N is

$$P_{\text{surcharge}} = 0.00082\ \mathrm{e}^{0.00289\left[(x-49.701)^2+(y-5.9187)^2\right]},\quad P_{\text{cut}} = 0.46$$

where P_{cut} is constant over the cutting region.

The error in the pressure models caused by the uncertainty in the sensor position was found in the same way as for the quadratic models to be $\delta P/P = 0.00041(x + y - 55.620)$.

The quadratic model calculates low decreasing pressures at the upper corners of the blade. This suggests that the quadratic model is a better representation of the surcharge distribution than the exponential model. Both the quadratic and exponential models calculate high pressures in the middle of the blade that decrease from the bottom toward the top of the blade, as well as outward from the middle, as expected. However, partway from the middle of the blade, the pressures begin to increase toward the outer edge of the blade in both models. This characteristic was not expected because of the shape of the surcharge on the blade; this increase could be because of the way the surcharge region extends beyond the edge of the blade. Overall, these models seem to capture the surcharge distribution for the 30.48-cm blade in JSC-1A with a cut depth of 0.635 cm; they are a good start for further distribution modeling, as well as for evaluating potential blade designs using FEA software.

The output from the quadratic and exponential models is less spatially variable than the measurement or point model. If the blade is not stiff, the higher variations in the point model could create local stresses from moments at connections between the blade and frame that would not be present when the smoother models are used for loading. The smoother models are more convenient to use in FEA, and they are conservative being based on the upper-limit of error, but the point-pressure model might provide different local results depending on the stiffness of the blade.

The distributed pressure models are scenario-specific. They are only valid for the base tests performed with the 30.48-cm flat blade in JSC-1A using a cut depth of 0.635 cm. These models are limited to the scenario when the surcharge just reaches the top of the blade. They are not valid for steady-state excavation, and no research has been done to determine whether they can be scaled up for analyzing

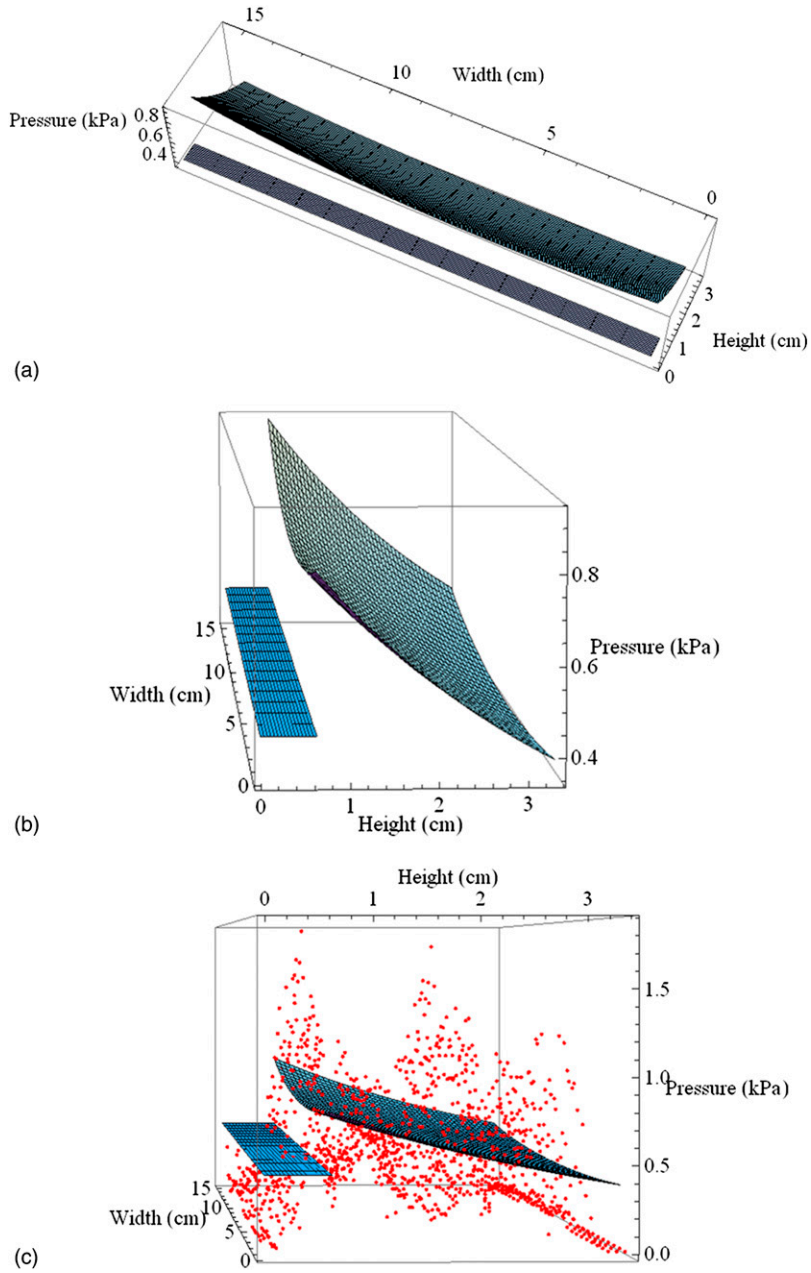

Fig. 19. (a) Exponential pressure model for base test scenario for 4.74-N target load with blade center at width = 0; cut height at 0.635 cm; (b) side profile; (c) overlay of point-pressure model data represented by dots

a full-size excavator blade. Additional modifications are needed for different soil types, because the surcharge might build up differently depending on the soil's coefficient of friction.

Conclusions and Recommendations

Considering the errors in measurement, the average pressure and load cell data are in agreement. The measurements showed that a V-shaped blade with an angle of 30° is the best choice, and the blade face of the excavator should be curved to minimize the cutting forces. These tests also suggest that multiple passes of small-depth cuts are the best excavation operating procedure to maintain low force and low power with a light-weight machine. The weight of the excavator blade can be minimized by employing a tapered curve along the blade and removing excess material from the support members.

The pressure distribution in Fig. 17 shows waves that are short in the vertical direction and longer horizontally. This is an interesting result that should be investigated further.

Three types of distributed pressure models were developed of the base tests with the 30.48-cm flat blade at a cut depth of 0.635 cm in JSC-1A with the surcharge at the top of the blade. While it has not been shown that they can be scaled up to full size, they can be used for load input to analyze small-scale blade designs with FEA software. One model is an array of point pressures for direct input into any FEA program, the second is quadratic for easy input into SolidWorks Simulation, and the third is exponential that is somewhat more precise for input into more robust FEA programs.

Future work could generate distributed pressure models for the remaining test scenarios and perform additional analyses with the data collected to develop models that predict effects of variations in cutting depth, blade width, blade shape, and rake angle. This

work should also include development of a time-dependent model to assess the effects of the rising surcharge pile. Further work is recommended to determine if the small-scale pressures can be scaled up to larger blades. Work should also be done to evaluate a range of extraterrestrial environmental conditions including low gravity, partial vacuum, extreme temperatures, and soil simulants.

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